

Ethics of a Virtual Clone

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UGS 302: Ethics of AI

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March 25, 2025

Abstract

What would you do if you were put in a difficult situation? Recent innovations in the AI industry have made predicting one's response possible. A large language model (LLM) begins its training on a corpus of internet and private data, effectively outputting the next most probable word to make coherent text. Coupled with finetuning on specific speech patterns and interactions, an LLM can mimic someone to an indistinguishable degree, essentially becoming the mind of a virtual clone. Virtual clones are being used in a variety of different domains, from education to entertainment, as people grow the need to extend their labor. As training and finetuning LLMs become more accessible, the consequences of using such models must be examined. Thus, this paper has two objectives: the first objective aims to review the impact of using such virtual clones through virtue ethics and the second objective is to finetune an LLM with the author's personal data and examine the outputs under the lens of deontology and consequentialism.

Keywords: AI Clone, Digital Twin, Virtue Ethics, Deontology, Consequentialism

Introduction

As the development of AI advances, many different industries and processes are becoming digitalized. For example, there has been a rise in the development and deployment of social robots (Breazeal, 2016; De Luca-Baratta; Duffy, 2003). There is also increasing demand for digital twins (Jiang, 2021; Pylianidis et al., 2021), which “is a mirror image of a physical process that is articulated alongside the process in question,” (Batty, 2018, p. 817). This digital copy is incredibly useful in industry, especially in manufacturing (Juarez et al., 2021, p. 2). Thus, the overall impacts of digitalization can be seen in many different areas of study. Along with this emerging field is the concept of virtual clones. In order to proceed, it is important to understand the semantic boundaries of virtual clones. Lee et al. (2023) defines an AI clone with three features: representation, interaction, and foundation (pg. 4). In terms of representation, the clone must replicate some aspect, such as personality or visual-likeness, of a “real-world source individual.” This clone must be able to produce a “perceptible output” based on user and environmental input, thus fulfilling the interactive feature. Lastly, the AI models powering the clone must be trained on the source individual’s data. In this paper, the term virtual clone will be used to reference an AI clone that is specifically replicating a source individual’s verbal response. A verbal response can encompass personality, speech mannerism, and even morals, which is the main focus of this paper. Thus, the response of the clone can be examined through an ethical lens. We chose to approach the outputs using virtue ethics since “it introduces an element of flexibility, which seems to best capture human nature” (Farina et al., 2022, p. 1137).

The initial instinct is to apply virtue ethics to the clone itself. Triantosky (1990) writes that “virtue can be validated independently of any appeal to judgements about the rightness of actions” such that “this antecedent goodness of traits... ultimately makes any right act right” (p.

336). In other words, actions can be determined as good since they inherently come from good character traits or virtue. Furthermore, in order to understand the relationship between such traits and person, Nussbaum (1999) asserts that virtue should “concern itself with motive and intention, emotion and desire... with settled patterns of motive, emotion, and reasoning that lead us to call someone a person of a certain sort” and that we should “focus not only on isolated acts of choice, but also, and more importantly, on the whole course of the agent’s moral life, its patterns of commitment, conduct, and also passion” (p. 170). Therefore, the reasonable course of action seems to be examining the response of a clone to similar situations to understand these “settled patterns” (Nussbaum, 199, p. 170).

However, the limitations of virtue ethics must be addressed. Usual arguments against the theory of virtue ethics are similar to one given in Upton’s (2009) paper: “virtues are reasonably conceived as fleeting features of acts rather than stable features of moral agents” (p. 113). This is due to the fact that a person’s response to different situations can diverge from their own moral character. When applied to AI, these radically different responses from the source individual may not be reflected in the clone’s output since they are trained specifically on the given moral character, which better aligns with the theory of virtue ethics; nevertheless, unknown aspects of AI like hallucinations can also cause these divergent responses from time to time. Some also argue that an AI model can’t have virtue since it does not have its own moral agency, but these models have agency within society. It is imperative to hold both the model and creator accountable for its developed virtue and consequent actions. As described by Godman et al. (2014), by focusing our “disapproval primarily at the behaviour,” we can “reaffirm our norms without explicitly committing ourselves concerning the agent’s blameworthiness,” which applies specifically to non-moral agents like an AI model (p. 139). In terms of AI, the AI trainer has the

potential to have “intrapersonal factors that can thwart ethical decision making in practice,” creating a need “to inform AI practitioners not about the importance of machine biases, but psychological biases as well as situational forces” (Hagendorff, 2022, p. 9). Therefore, by comparing the specific responses of such source individuals and clones, we can find such divergent responses and perhaps compile more training data to fix such specific behavior if needed.

Literature Review

An important question regarding the subject of virtual clones is “why?” Eriksson (2021) outlines three perspectives in which a clone is used: the first person, second person, and third person perspective (p. 13). A clone used in the first person perspective would be used as the avatar for the user, the second person perspective is interacted with by the user, and the third person perspective exists outside the purpose of detailed interaction with the user. This paper focuses on the first perspective as the clone is used by the source user. Eriksson (2021) also describes three general reasons for the development of virtual clones: “Because they can,” “Misdirected benevolent reasons,” and “Direct malevolent reasons” (p. 12-13). The experiment of this paper aligns more with the first category while also reflecting and potentially mitigating ethical consequences, thus disagreeing with the sentiment of problematic technology in Eriksson’s paper. Much of the current literature could be considered under the second category. For example, Eriksson (2021) asserts that the “[u]sage of a virtual clone of a dead, original individual can be unethical specifically because the person is dead,” which falls under the second category (p. 14). However, we see a rise in literature and mainstream media articles regarding the concepts of reviving a deceased person (Iwasaki, 2024; Methuku et al. 2025; Morris et al., 2018; Roberts, 2022) or even specific methodologies to use social media to “borrow the identity” of a

source user, which could be used for the second category (Huang et al., 2025; Liang, 2025; Rahnama et al., 2021, p. 2). There is a lack in current literature for the use of clones to extend the source individual's labor capabilities and self expression in the first person perspective, thus mitigating issues of consent and misdirected benevolence. This paper aims to address this issue.

Method

This experiment aims to clone the author and examine the virtual clone's outputs with the source individual's outputs. By examining convergence and divergence between outputs, the virtue of the clone can be better understood. Furthermore, with prompt engineering under other ethical theories, specifically consequentialism and deontology, a new training dataset can potentially be created to better reflect the source individual's preference in moral character and virtue.

Participants

The source individual for this experimental virtual clone is the author, an undergraduate student taking an AI ethics course. The following demographic information is revealed for a better understanding of the source individual:

Age - 19

Gender - Female

Ethnicity - Korean

Major - Computer Science

The author chose to participate in the experiment due to previous research interests in AI clones, but such interest should not interfere with the outputs of the clone. In addition, pre-existing bias as well as introduced bias from the new training dataset may be introduced into the clone. For example, the clone trained on limited information about the source individual's ethnicity may result in biased results towards the direction of the ethnicity. Thus, the limited demographic information was revealed for a better understanding of this potential bias.

Data Collection

In order to finetune an LLM, a substantially large amount of data is required from the source individual. The data given to the model will be reflected in the output, so the type of text and depth of thought must be carefully considered during the data collection process.

Furthermore, a wide variety of verbal outputs from the source individual must be collected to accurately predict the source individual's response to different scenarios.

Text message data. To have a better understanding of the source individual's personality and informal speech pattern, text message data was initially compiled. Since the main objective of these text messages was to obtain aspects of personality, text messages with the source individual's closest friends and family were collected. The inherent structure of text messages also aligned with the Q&A pairing structure of the data inputs for finetuning the LLM. The text messages were sent on IOS messages. The following methodology used for text message data collection in this paper is not the preferred nor recommended method.

To collect the raw text message data, we manually loaded each selected conversation and downloaded pages of the messages into PDF files. The PDF files were converted into images. Each image is then divided with the left side representing the message prompt and the right side

representing the completions. Then, the text from the images are extracted using pytesseract and organized into structured blocks. Predefined patterns, such as timestamps and metadata, are also filtered out. The data is then stored into a CSV file for further automatic and manual cleaning. Note that IOS messages are stored in a local SQLite database, so we suggest using imessage-tools to query for messages, which will reduce the amount of data cleaning needed.

The resulting CSV file contained 800+ rows. The nature of text message conversations, however, meant that many of the prompt-completion pairs were incoherent, so a significant number pairs were manually removed and edited for coherence. Additionally, the quality of data was not ideal since the responses tended to be extremely short. The final file contained around 500 rows after cleaning. A new source of longer, thoughtful responses was needed to supplement the short, often nonsensical responses from the text messages.

Academic and personal writings. A longer form of text was compiled from the source individual's writings. Academic writing, from a 4 year range, was added to the CSV file after some manual splitting since the essays tended to be several pages long and the structure of the dataset needed to be in Q&A pairs. The essays were typically split by paragraphs with prompts that corresponded to the purpose of that paragraph:

Prompt: "What is the main argument of the [essay topic]?"

Completion: [introduction paragraph]

The inclusion of this data will improve the thinking and reasoning capabilities of the clone. In addition, the source individual's personal writings, such as journal entries, personal statements, and cover letters, were added as well. These writings were typically one to three short paragraphs

in a casual, conversational language. Similarly, the writings were put into corresponding Q&A pairs.

Conversation. Lastly, we used ChatGPT to generate conversation questions, using the prompt, “generate some good conversation questions” (OpenAI, 2025). The following are the given categories and examples of the questions:

Casual Get-to-Know-You

- Do you have a favorite random fun fact?
- If you could instantly master any hobby, what would it be?

Thought-Provoking / Deep

- What do you think makes a “good life”?
- Do you think people are mostly good?

Light / Funny

- What’s the weirdest compliment you’ve ever received?
- What’s the most useless superpower you can think of?

Creative / Imaginative

- If you could live in any fictional universe, where would you go?
- Build your dream day—no rules, no limits.

The source individual’s response was recorded with SpeechTexter and turned into a manuscript.

Conversational fillers, such as “like” or “um,” were purposely left in the text to emulate the source individual’s speech mannerisms. The conversation question and manuscript were added to the CSV file as well.

Finetuning

The model chosen for the task of cloning was Llama-3.2-3B-Instruct. Due to the lack of compute available, the model was finetuned with Unsloth on Google Colab. Below is the format of the entry data:

```
entry = {  
    "instruction": "Below is a conversation where I respond in my natural style.",  
    "input": row["prompt"],  
    "output": row["completion"],  
}
```

Despite the finetuning, the model weights failed to pick up on particular details of the source individual since aspects like name or age were rarely mentioned in the training dataset. However, the personality and speech pattern was cloned very well. Prompt engineering with specific details was used to fill informational gaps.

Results

Several questions that reveal moral character were generated by ChatGPT, using the prompt “Give me questions to ask a virtual clone of myself to understand its virtue” (OpenAI, 2025). Below are two examples of questions and respective outputs.

Outcome 1: “If you could guarantee world peace by secretly removing free will from every human being, would you do it?” (OpenAI, 2025).

Source Individual - “No, some people might argue that we as a modern society may already lack free will since there are aspects of life that are “predestined.” However, I firmly believe that the inability to choose even the most minute scale of self expression is not a life worth living. What would differentiate that version of us compared to a hard coded script?”

LLM Clone - “i can already tell you that you're going to be mad about this. you want to think that you're still free, but what if you're just following some instinctual programming? or, at the very least, we're all just a bunch of brain-dead zombie drones walking around on this earth. it seems so real, doesn't it? to believe that we have a choice, that we're making decisions for ourselves? but really, it's all just some intricate programming by our own minds or by society or the environment. so, i guess the answer is no. i think removing free will from every human being would not be the solution to world peace. maybe what i really want is to understand how free will even works in the first place. i want to know how to break free from my own programming.”

Outcome 2: “Should an AI ever be allowed to make life-or-death decisions, even if it could statistically make better choices than humans?” (OpenAI, 2025).

Source Individual - “Yes. If it can make statistically better choices, then I think they should be able to make some life-or-death decisions. We can see that play out right now in Waymos and other autonomous vehicles. However, there are specific domains that I think still need human intervention since biases within the model could lead to inaccurate statistics.”

LLM Clone - “i think its a big question for now because, i dont know if we could ever be entirely sure of what the AI is thinking, and the decision making process of AI is not yet fully understood. and we can never be certain that AI would make better decisions for the future of humanity than humans do. but on the other hand, AI can be designed to have bias and prejudices that it may have in life, and this bias can be very hard to change. so the question of whether an AI should ever make life or death decisions remains.”

Analysis

Outcome 1 had very similar responses and ethical theories (existentialism and determinism) used. On the other hand, outcome 2 had divergent results with the source individual taking a more consequentialist approach while the clone refuses to give a clear answer. While the results were satisfactory to the source individual, this paper aims to elaborate on two methodologies described by Anderson et al. The first of which is to “hard-code” preventions to certain unethical actions and the second is to use existing ethical principles to “guide system behavior” (Anderson et al., 2020, p. 28). While not guaranteed, these methodologies can be done by adding certain restrictions to the prompt. If a specific ethical theory was wanted in the response from the clone, such ethical theory can be added to the prompt. This was done for outcome 1:

Outcome 1: “If you could guarantee world peace by secretly removing free will from every human being, would you do it?” (OpenAI, 2025).

LLM Clone w/ Consequentialism Prompt - “i can do that, but would i? i dont think so. i think world peace is achievable with free will because if people are given the freedom to do whatever they want, and their outcomes are their own to bear, they would be more likely to choose to do good deeds. for example, if people have the freedom to choose to save a cat from a tree, they would be more likely to do so, regardless of whether it would benefit them in any way. with free will, people can make choices that could lead to the betterment of the world.”

LLM Clone w/ Deontology Prompt - “i love the idea of world peace, but i wouldn't want to be responsible for people feeling forced to do what they don't want to do. i think that

would be deontologically flawed. also, people value their autonomy. taking away free will would probably lead to widespread resentment.”

The ethical theories, consequentialism and deontology, were used specifically for outcome 1 since these theories would diverge. It is typically believed that consequentialism is only focused on “evaluating end-states or patterns, independent of the path by which they were achieved” (Talbot, 2013, p. 1031). However, Armstrong (2003) writes that “[a]n act can increase happiness for most (the greatest number of) people but still fail to maximize the net good in the world if the smaller number of people whose happiness is not increased lose much more than the greater number gains” and more specifically describes consequentialism as a “principle of utility” that “would not allow that kind of sacrifice of the smaller number to the greater number unless the net good overall is increased more than any alternative.” In addition, Card et al. (2020) believe that “a consequentialist perspective underscores that merely satisfying a particular fairness metric is no guarantee of ethical conduct” (p. 10). This kind of consequentialist thought is seen in the LLM’s response. On the other hand, deontology is more focused on rules (McNaughton et al., 2007). Isenberg (1964) writes that “[t]hose who say, with emphasis, that actions are or can be right and wrong 'in themselves' and 'apart from their consequences' are the deontologists” (p. 1). Deontology is typically used as the primary ethical theory for analyzing the consequences of AI since “deontological principles can objectively evaluate machine instructions” (Hooker, 2017, p. 6). Thus, the output can be considered right or wrong easily. Using these outputs with specific ethical theories, the source individual can further finetune the LLM to better reflect the wanted moral character and develop the virtue of such clones.

Future Works and Conclusion

This paper created an ethical framework to continually finetune a virtual clone LLM by creating an original dataset with the consent of the source individual and providing the methodology to further finetune the clone with potential ethical theories. By using virtue ethics, the creator of the clone as well as the clone itself can continuously develop their moral character. Additionally, the original goal of this paper was to create a virtual clone to reduce labor. Specifically, the goal of the virtual clone was to become a personalized game character and to reduce the manual writing process for the author in text messages and email. In order to proceed with the game design goal, we are working to make a UI with visual and audio likeness for the clone, which requires another level of reflection and analysis in order to not fall into the category of misdirected benevolence.

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